



UNIVERSITY OF AMSTERDAM

Alternatives to Attribution & Omnichannel Management

Social Media & Social Media Analytics

Introduction to Unstructured Data: Text, Photo and Video Data

Digital Marketing and Analytics: Week 2

Academic Year 2025-2026 – Semester 1 – Block 2

AMSTERDAM
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Business

Challenges & Alternatives Attribution

Challenges in Attribution 1: Varying-Roles of Touchpoints

A TOUCHPOINT can have three roles in a customer journey— **INTRODUCTION, ASSIST, AND CONVERSION**

Optimal mix of TOUCHPOINTS that provide the highest return on investment (ROI) – while being **used in the right stage (with the right timing)**.

EXAMPLE: What if a Touchpoint A is very influential on customer purchase ONLY when it is used in Consideration stage and not in other stages. So the problem is NOT only – WHICH touchpoint – but WHEN and HOW ?



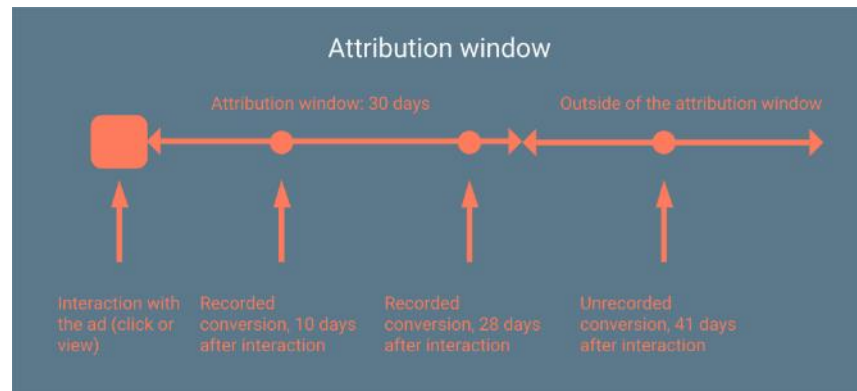
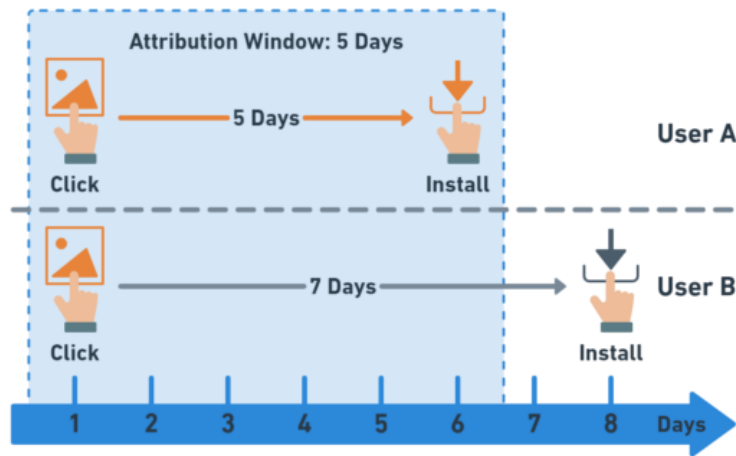
Source: <https://www.thinkwithgoogle.com/tools/customer-journey-to-online-purchase.html#/the-us/travel/medium/generic-paid-search>

Challenges in Attribution 2: Time-Window

Our window ends at Day 6: WHAT IF the journey is NOT complete yet – and the customer comes back and purchases at Day 7. Note: Taking Day 6 as the end of our **Attribution Window** we would have assumed that it was a journey ending with no-purchase (no conversion). But in reality: It is NOT.

SOLUTION: No definite solution since the customer may always come back to purchase. However **statistics on average customer journey durations for different industries** may be helpful in deciding on the most accurate attribution window (time-slot).

If you have an online store, most conversions will take place 7-10 days after a user first visits your site: Know your (approximate) sales cycle



Challenges in Attribution 3: What If Customer Did Not Click (But Saw the Ad)

VIEW-THROUGH ATTRIBUTION: What if the customer did NOT click – but only saw your ad / or content ?

According to ComScore, only 16% of Internet users click on display ads. Thus, the impact of display ads on the remaining 84% remains unknown. Furthermore, the number of Internet users who don't click on ads increases each year.

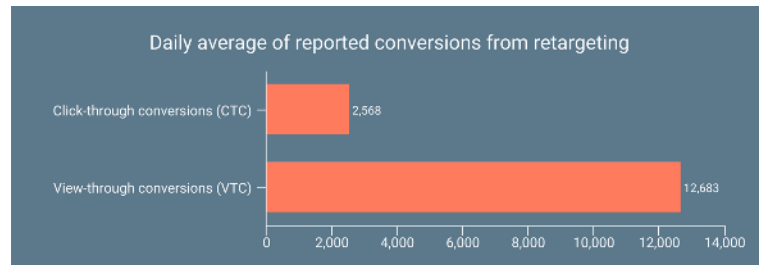
Heavy, Moderate, Light Display Ad Clicker Analysis

March 2009 vs. July 2007

Total U.S. – Home/Work/University Locations

Source: comScore

	Share of All Internet Users		Share of Click-Throughs	
	Jul-07	Mar-09	Jul-07	Mar-09
Total Clickers	32%	16%	100%	100%
Heavy Clickers	6%	4%	50%	67%
Moderate Clickers	10%	4%	30%	18%
Light Clickers	16%	8%	20%	15%
Non-Clickers	68%	84%	0%	0%



Challenges in Attribution 3: What If Customer Did Not Click (But Saw the Ad)

The ratio of click-through conversions to **view-through conversions** is approximately 1:5. However, counting view-through conversions is difficult.

A/B test could be helpful in order to see the effect of views on conversion = View-Through-Conversions

Audience is divided into two groups.

The first group: an ad for the store with a reminder about those products that had interested them earlier.

The second group: saw an ad that was completely unrelated to the store that they had visited earlier.

Increase in conversions = (conversion rate of AC №1 – conversion rate of AC №2) / conversion rate of AC №2,

Increase in conversions = (0,304 – 0,241) / 0,241 = 26% (through view-through; not clicked but supposedly seen ad)

Campaign	Unique users	Total conversions	View-through conversions	View-through conversion rate	Increase in conversions
Store ad (campaign №1)	127,163	413	386	0.304%	26%
Ad unrelated to the store (campaign №2)	104,883	256	253	0.241%	

Challenges in Attribution 4: GDPR – Data Privacy & Protection

GDPR and other legislations makes it less possible and complicated to directly reach and use clickstream-session data on **customer touchpoint exposures** which is necessary for Attribution Modeling-Analytics.

Now there are more and strict rules on the use of customer touchpoint exposures for analytical purposes.

For Example: Shorter tracking times (no longer possible to track customer journeys for weeks) – Then what is the alternative?



GDPR PERSONAL DATA

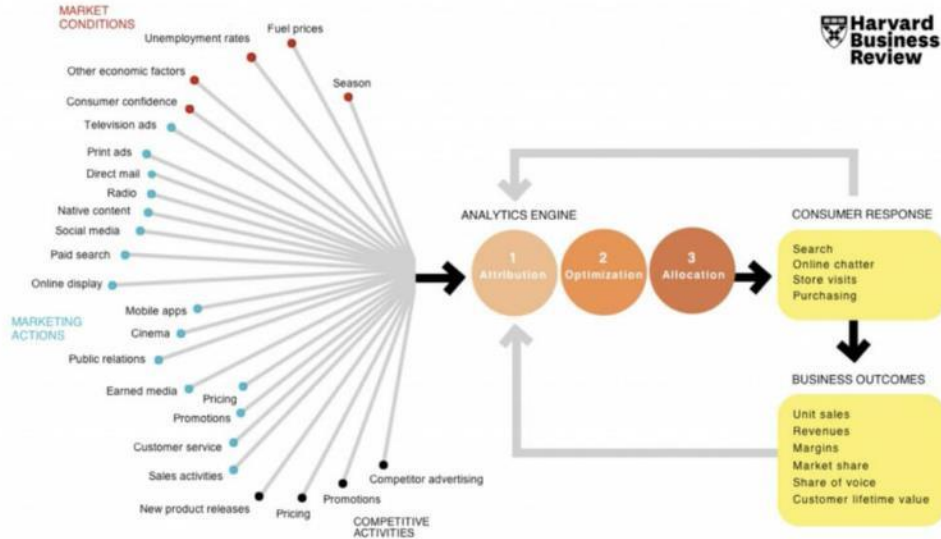
The EU's General Data Protection Regulation defines personal data as any information related to a person that can be used to directly or indirectly identify them, including:



Media (Marketing) Mix Models (MMM): Alternative to Attribution

MMM is a marketing analysis technique that measures the impact of various marketing efforts on sales; by incorporating **various components in the marketing-media mix** and using regression models.

Unlike Attribution; **MMM** do NOT ONLY focus-use Touchpoint Exposures and incorporates other non-touchpoint related effects (such as environmental factors, customer service, distribution and even competition efforts). Thus a wider range of factors.



Besides complex equations, the MMM consists of :

1. What marketing instruments are you using?
2. How much money are you spending on each instrument?
3. What were previous campaign results and insights?
4. What other factors may impact your sales/ Are they measurable?

Media (Marketing) Mix Models (MMM): Alternative to Attribution

MMM is on the rise mainly because (i) death of cookies (ii) restrictions on online-touchpoint tracking (iii) by MMM you can consider and include more elements in your analysis – even weather or seasonal factors.

Here is a sample list of variables that could be used as **Independent (Explanatory)** or **Dependent (Outcome)** Variable in an MMM Model.

Dependent variable

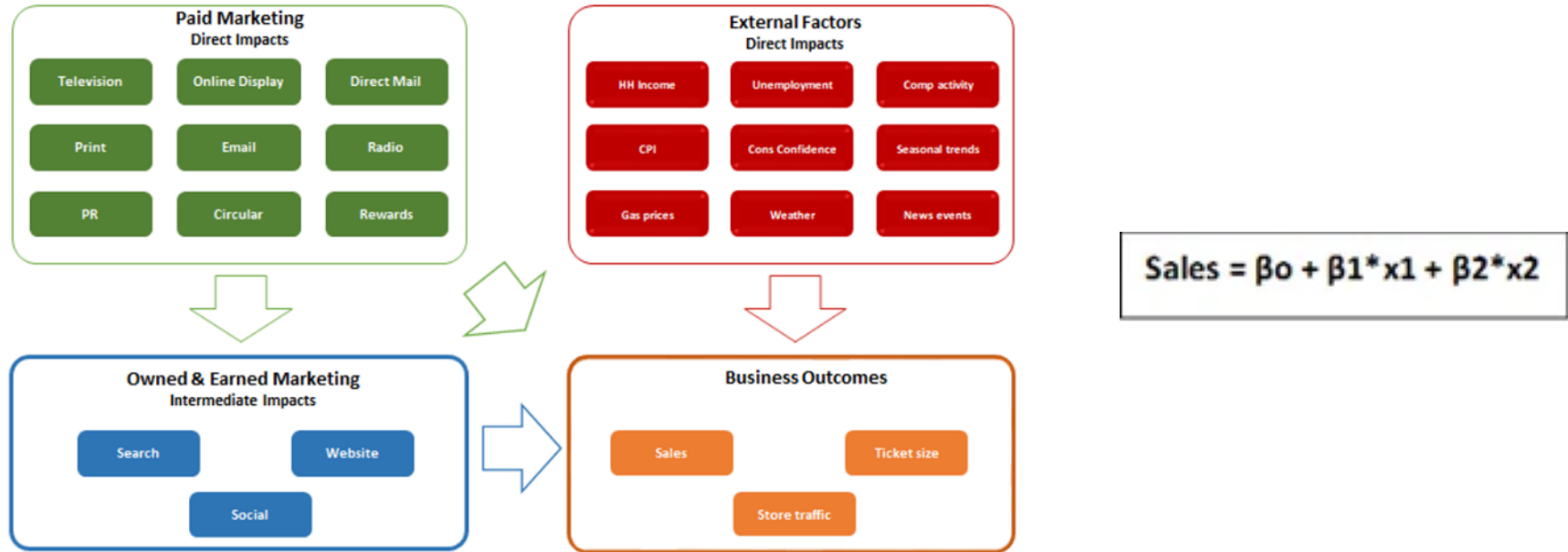
	The dependent variable in MMM is the variable that you are trying to measure.
	Sales Total amount of money that a business generates from its products or services.
	Revenue Total amount of money that a business generates after taking discounts and returns.
	Customer acquisition cost Average amount of money that a business spends to acquire a new customer.
	Customer lifetime value Total amount that a business expects to generate from a customer over their lifetime.
	Product satisfaction Measures how satisfied customers are with a product or service.
	Market share Measures the percentage of the total market that a business controls.

Independent variable

	Independent variables in MMM are the variables that you believe are influencing the dependent variable.
	Advertising spend Amount of money invested in promoting a product or brand through various channels.
	Price Includes the price of a product or service, as well as any discounts and promotions.
	Product features Includes the features and benefits of a product or service.
	Competitor activity The marketing activities that a business's competitors are doing.
	Economic conditions Overall economic health of the market that a business operates in.
	Seasonality Seasonal demand changes, such as holiday spikes.

Media (Marketing) Mix Models (MMM): Alternative to Attribution

Purpose of using MMM is to understand how much each marketing input contributes to sales, and how much to spend on each marketing input. MMM uses the Regression technique for extracting key information/insights.



Media (Marketing) Mix Models (MMM): Alternative to Attribution

MMM and **Attribution Modeling** are compared with their advantages and disadvantages in the table below.

Multi-touch attribution



- ✓ Quick and easy to implement
- ✓ Tracks individual touchpoints
- ✓ Quick insights for real-time optimisation
- ✓ Holistic view of customer journey



- ✗ Limited to tracking click path data
- ✗ Less effective due to the death of cookies
- ✗ Can overvalue demand-capturing channels
- ✗ Focuses on short-term activity

Marketing mix modelling



- ✓ Quantifies everything, including the weather
- ✓ Considers invisible touchpoints (ad views, TV)
- ✓ Privacy-safe—doesn't rely on cookies
- ✓ Can demonstrate diminishing returns

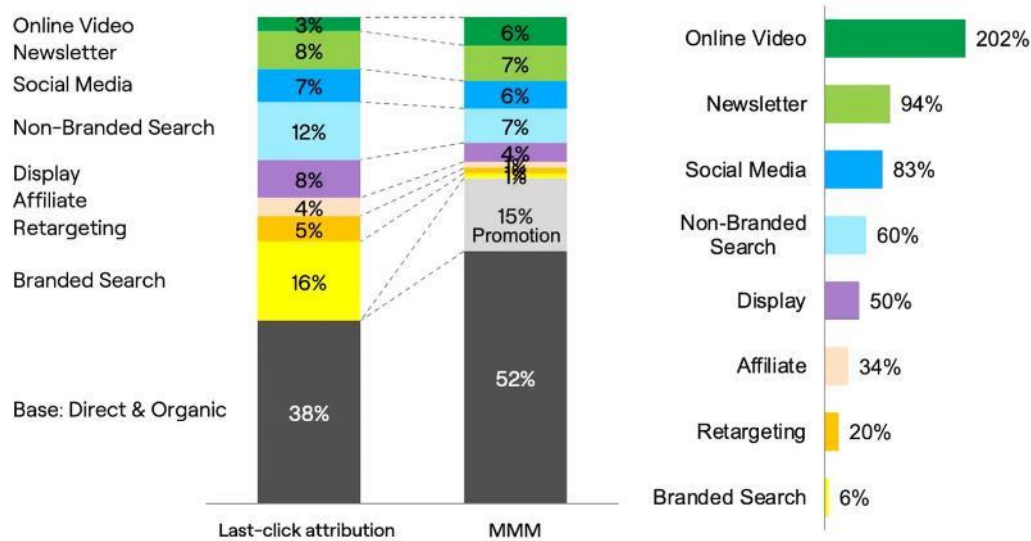


- ✗ Can't show individual touchpoints
- ✗ Can take between 3-6 months for analysis
- ✗ Requires a large amount of data
- ✗ Expensive and time-consuming to build

Media (Marketing) Mix Models (MMM): Example – Comparative Results

See how the results of **MMM** and **Attribution Model** may differ in the example below:

MMM also reveals a strong impact of **Promotion** activity on Sales – which is invisible in Attribution Model results (that only focuses on touchpoints)

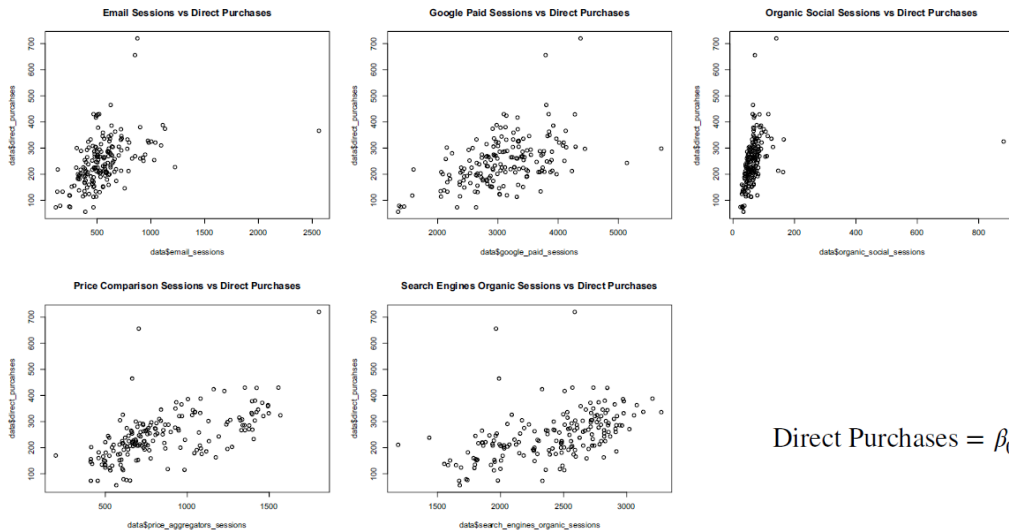


Simplistic Inference: Linking Web Traffic (Channels) to Online Purchases

Google Analytics (GA-4) can also be of help to explain online purchases – to some extent (if no other data is available)

Research finds there is an **association** between website traffic from various **digital marketing channels** and direct **online purchases** (from website)

Use/Exposure to three marketing channels explain 61% of the variation in online sales coming from direct website visits-purchases.



3 channels explaining 61% of Online (Direct) Purchases

- **Google (Paid) Ads,**
- **Price comparison engines**
- **Email marketing**

$$\text{Direct Purchases} = \beta_0 + \beta_1 \text{Google Paid} + \beta_2 \text{Price Comparison} + \beta_3 \text{Email} + \varepsilon$$

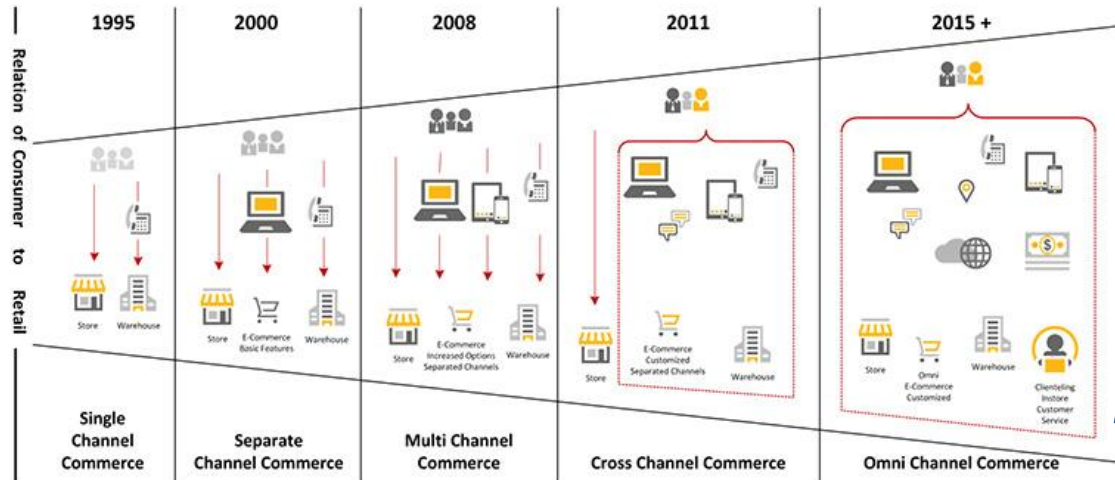
Omnichannel Management

Online-Offline & Cross-Channel Interplay

Multi-Channel to Omnichannel Management

Marketing landscape evolved from a perspective

From where **channels seen as separate** (multichannel) to a perspective where they are seen as **integrated and in an inter-play** with each other



OMNICHANNEL VS MULTICHANNEL

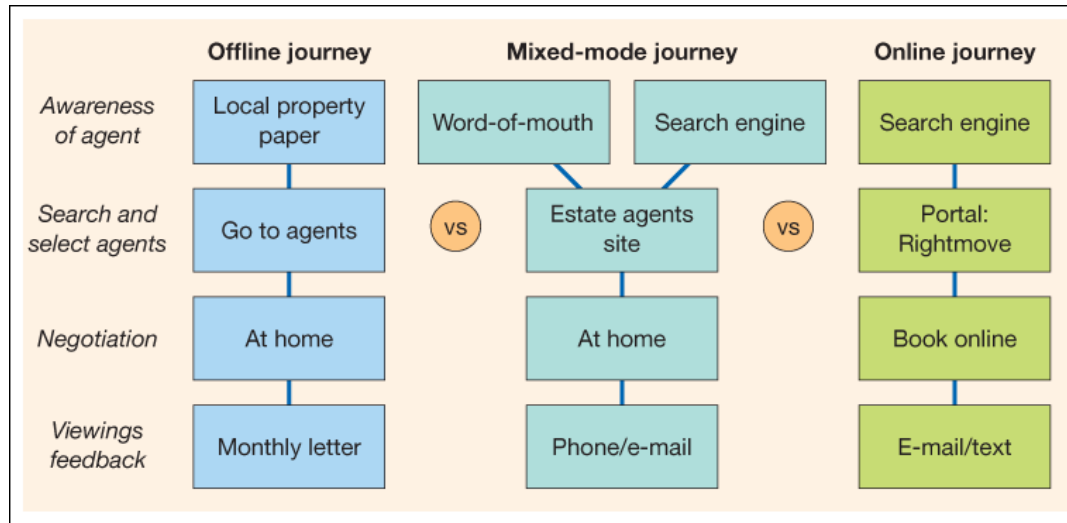
"What is different in Omnichannel"

- Channels Interact/ Link with each other.
- Channels co-exist (operate) in omni-presence
- Seamless Customer Experience
- Touchpoints are important as channels
- Holistic Experience & Moments has central role

Online, Offline and Mixed-Mode Journeys

Example of a channel chain map for consumers selecting an estate agents to sell their property

Do you think (apart from mixed-mode journeys) **offline-only** or **online-only** journeys exist today ? In small or big percentages % ?



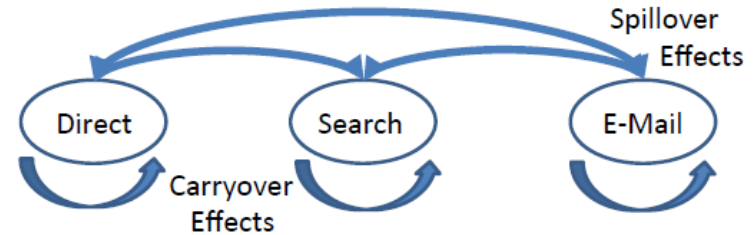
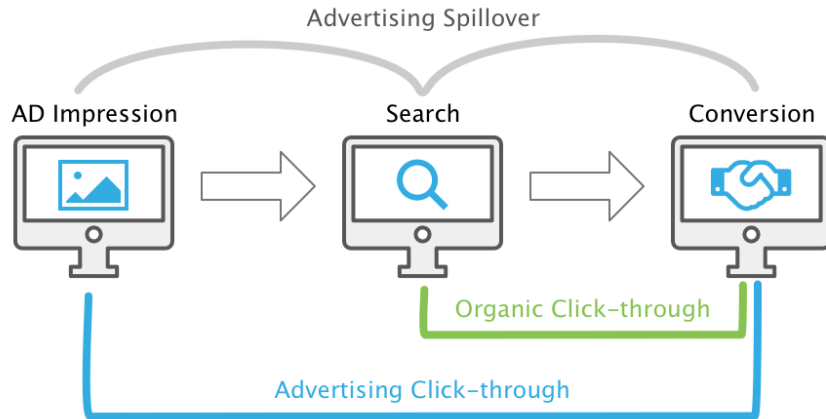
Cross-Channel Effects: Carry-Over & Spill-Over Effects

CARRY OVER EFFECTS: The effect of a channel on the same channel in the subsequent (later) phases

Example: Facebook ads (that you liked) will have positive impact on you clicking Facebook ads again (later) !

SPILL OVER EFFECTS: The effects of a channel on another channel (cross-effect)

Example: The emails you clicked (and read) direct you to a social media ad that you were not aware of !



Source: Visuals > VENDASTA company website: <https://www.vendasta.com/blog/digital-advertising-spillover-organic-seo>
Li, H., & Kannan, P. K. (2014). Attributing conversions in a multichannel online marketing environment: An empirical model and a field experiment. *Journal of Marketing Research*, 51(1), 40-56.

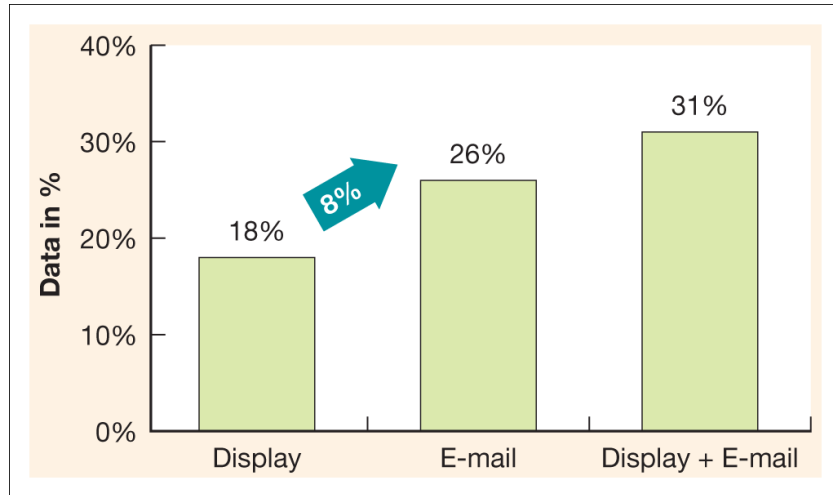
Cross-Effects: Touchpoint Synergies

Combined use of touchpoints can (possibly) lead to synergetic outcomes: see the figure

If we observe a **higher purchase (conversion) %** in the case of a combined use of two touchpoints: we can talk about **synergetic effects**

Can you also think of **cannibalistic** cross-channel effects ? (certain channels – if used together- have negative effect or making each other less visible?)

Note: You can analyze and reveal cross-channel effects by using Interactions in Regression (Tutorial 1)



Online-Offline Interplay: Research Shopping

Research Shopping: When a customer uses different channels to search and purchase the product.

Online Search > Offline Purchase: Webrooming (most common type 5-10 years ago, it still is)

Offline Search > Online Purchase: Showrooming (Many people still prefer seeing and touching the merchandise they buy before they buy online)

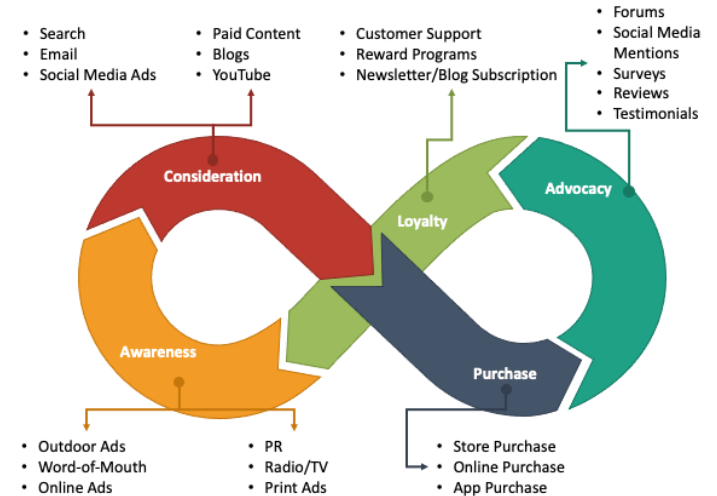
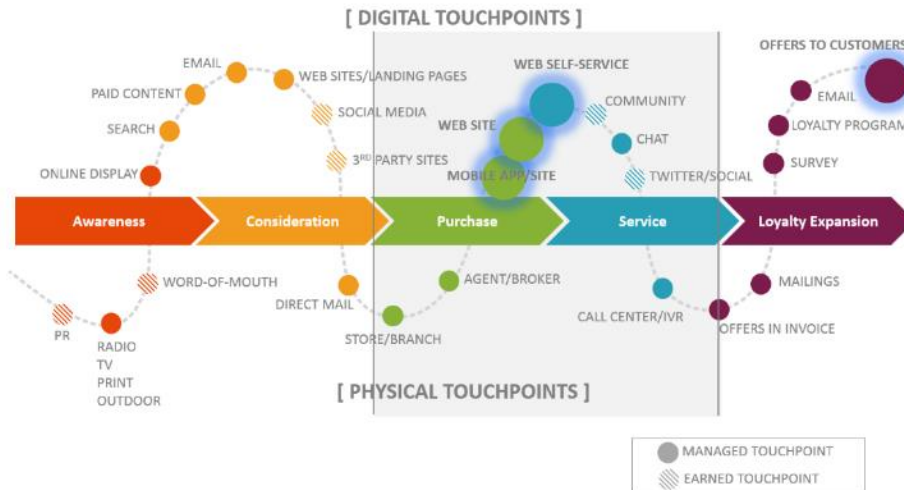
		Research...	
		Online	Offline
Mobile buyer	Online	22%	9%
	Offline	37%	32%



Customer Journeys in Omni-Channel World

Customer Journey (maps) visualize multiple stages of customer journey in more detail.

Customer Journey is **rather a cycle-loop** rather than **straight timeline**: they repeats in an evolving manner, being updated with new experiences



Online-Offline Customer Journey Maps

A diagram that depicts the stages that customers go through in interacting with a company, from search to transaction and aftersales



Fig. 4. Hypothesised Customer Journey for Health Care Insurance Service.

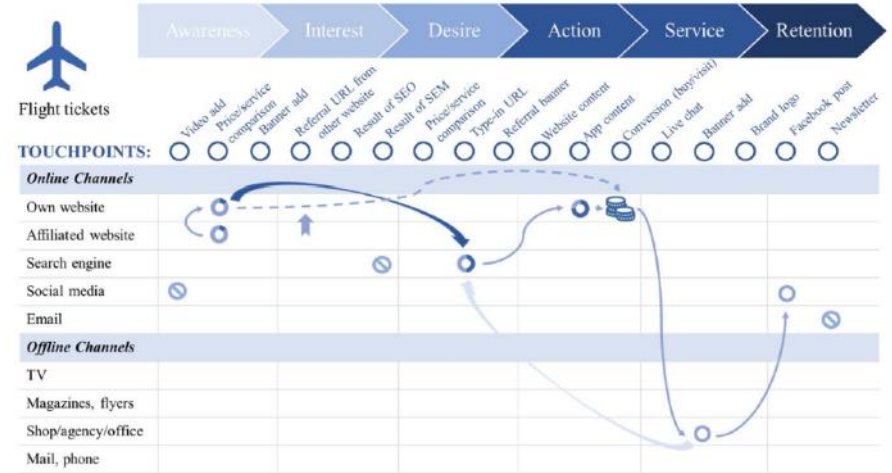


Fig. 5. Hypothesised Customer Journey for the Low-Cost Airline Tickets.

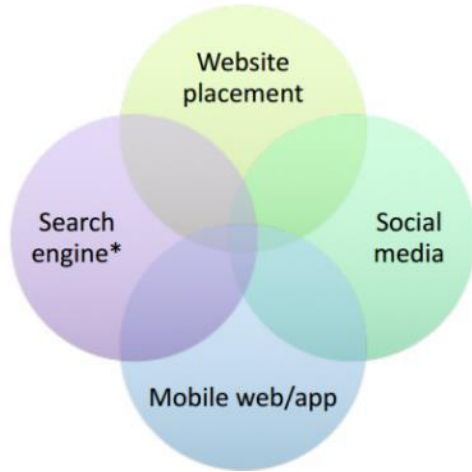
Introduction

Social Media Marketing & Analytics

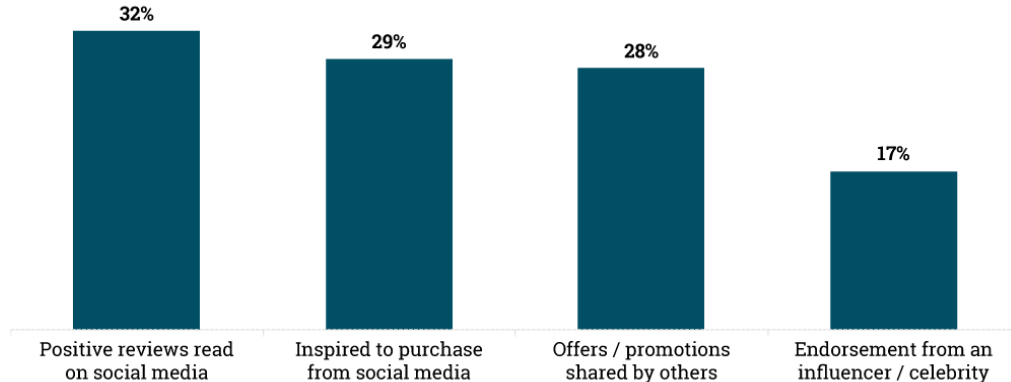
Social Media: New Dominating Channel in Customer Journeys

One of the four main-pillars in online marketing:

*"Has any interaction you have had with your favorite brands on **social media** driven you to buy more?"* > **80 % YES** (Global PwC Survey)



Social Media's Influence on Consumer Purchase Decisions (% of global respondents reporting influence)



Published on MarketingCharts.com in April 2019 | Data Source: PwC

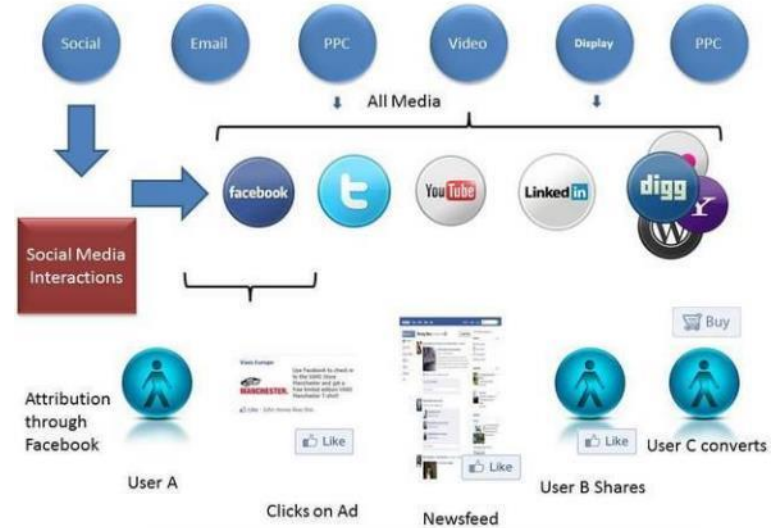
Based on a survey of more than 21,100 online consumers in 27 territories

代言/背书



Social Media: Role(s) in Customer Journey

The relative share of Social Media Platforms (as touchpoints) and their attribution value (impact on purchase) is on the rise in the last decade.



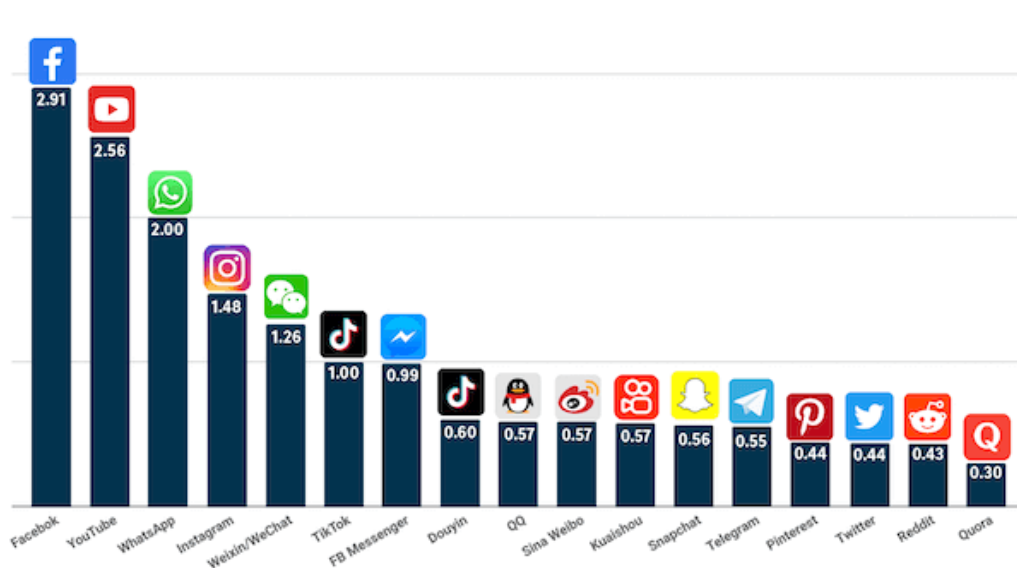
Social Media: Usage Patterns Differ per Platform

Social Platforms has their own patterns, appeal and user audience.

Can you guess: What is the social media platform with the highest (%) Conversion Rate (from direct links to purchase)

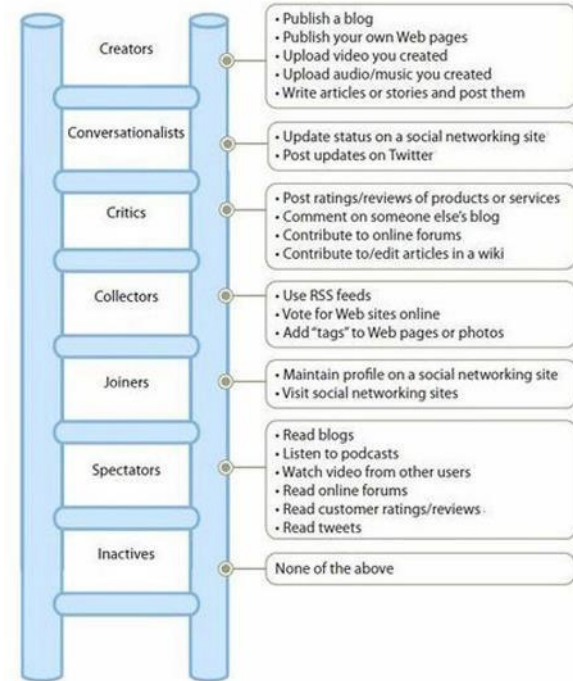
Can you guess: What is the % of active twitter users ever typed/posted a tweet ?

Let's check: I guess everyone here is using Youtube, or Instagram – Who is using Tiktok, Anyone using Discord or Clubhouse ?



Social Media: Levels of Customer Engagement

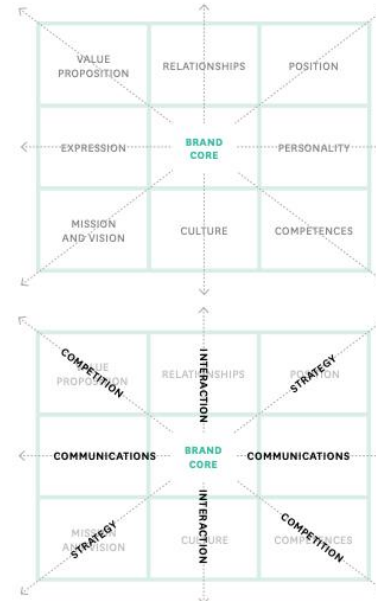
What people are doing



Social media and Corporate Brand Identity

A corporation's identity is made up of nine interrelated components. By examining each one and how it relates to the others, an organization can build a stronger brand. *“Could you to create a matrix based on where you currently work or would like to work in the future?”*

EXTERNAL/ INTERNAL	EXTERNAL	VALUE PROPOSITION What are our key offerings, and how do we want them to appeal to customers and other stakeholders?	RELATIONSHIPS What should be the nature of our relationships with key customers and other stakeholders?	POSITION What is our intended position in the market and in the hearts and minds of key customers and other stakeholders?
	EXTERNAL/INTERNAL	EXPRESSION What is distinctive about the way we communicate and express ourselves and makes it possible to recognize us at a distance?	BRAND CORE What do we promise, and what are the core values that sum up what our brand stands for?	PERSONALITY What combination of human characteristics or qualities forms our corporate character?
	INTERNAL	MISSION AND VISION What engages us (mission)? What is our direction and inspiration (vision)?	CULTURE What are our attitudes, and how do we work and behave?	COMPETENCES What are we particularly good at, and what makes us better than the competition?



Source: Harvard Business Review (2019) <https://icc.se/wp-content/uploads/2022/03/Urde-Greyser-2019-HBR-What-Does-Your-Corporate-Brand-Stand-For.pdf>

Enhancing the Impact: Influence & Influencers

What Drives Influence (Influence Model):

The probability that a person (I) choose potential influencer A over B will increase with the number other people choose A (social proof)

1) Threshold rule 2) Network 3) Dynamics

1) 阈值规则 2) 网络 3) 动态



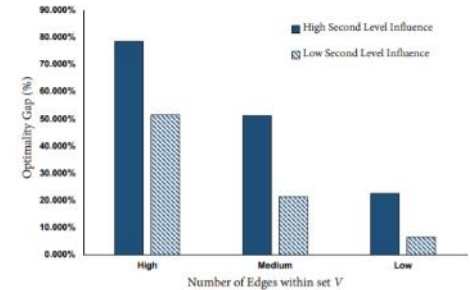
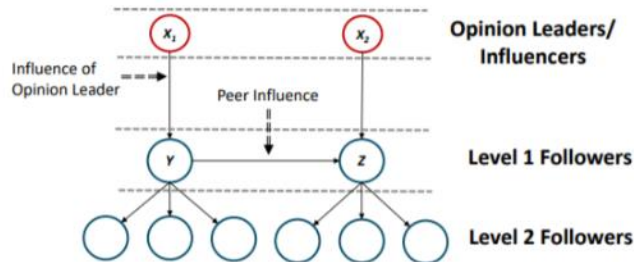
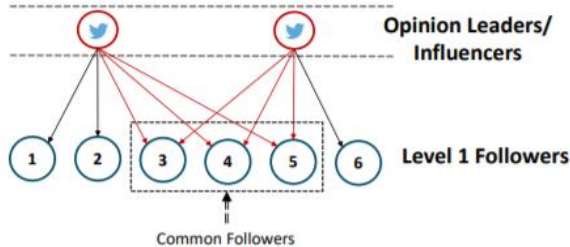
Source: Peters, K., Chen, Y., Kaplan, A. M., Ognibeni, B., & Pauwels, K. (2013). Social media metrics—A framework and guidelines for managing social media. *Journal of interactive marketing*, 27(4), 281-298.

Enhancing the Impact: Influence & Influencers

To determine the influence power – better to look at not only Level 1 followers but Level (k) followers: followers of followers:

Thus two main parameters to determine the influencers:

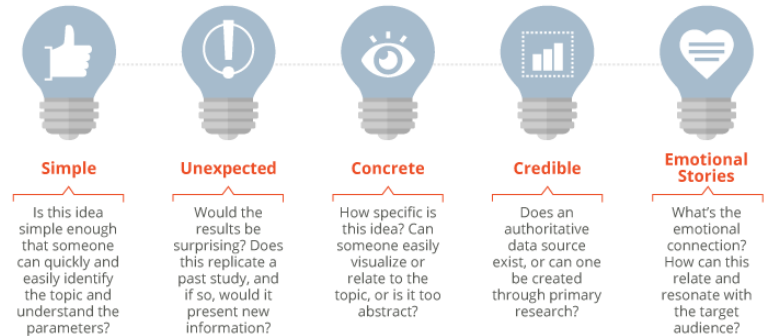
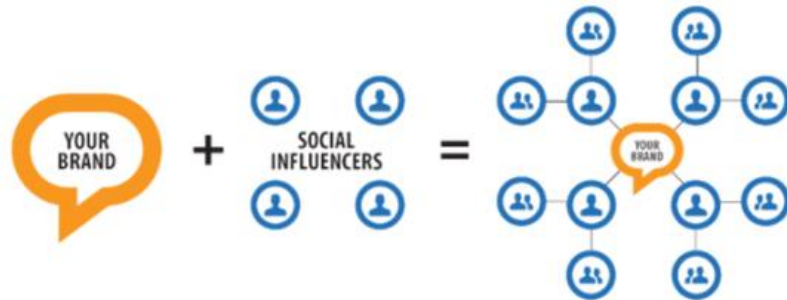
- 1 Number of K (chain) level followers
- 2 Likelihood (activation) level of each $(1+...+K)$ level follower to share the content



Social Media: Enhancing the Impact: (2) Virality

Virality: the likelihood of something being shared and spread widely among different users, or customers online.

Things spread or go viral: (1) Only if recent (2) Only rare events. **BUT can we predict and/or create-manage VIRALITY?**



A dark blue rectangular box containing the formula for the Virality Rate Percentage. It shows 'Shares' over 'Impressions' multiplied by 100 equals 'Virality Rate Percentage'. A small owl icon is in the bottom right corner.

$$\frac{\text{Shares}}{\text{Impressions}} \times 100 = \text{Virality Rate Percentage}$$

Source Visuals: 2xceed company website: <https://2xceed.nl/influencer-marketing/>

Source: Hootsuite company website (formula visual)

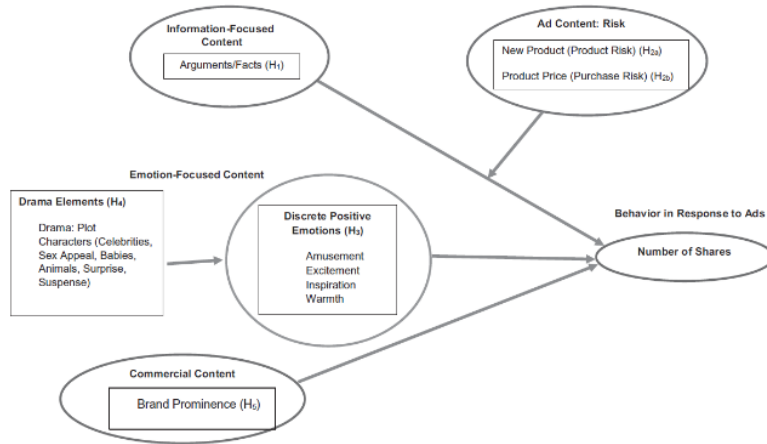
Virality: What Makes an Online Content Go Viral ?

Positive content drives virality.

High-Emotional Arousal: positive (awe) or negative (anger or anxiety) emotions is more viral.

Positive emotions of amusement, excitement, inspiration, and warmth positively affect sharing.

Drama elements such as **surprise, plot, and characters, including babies, animals, and celebrities** arouse emotions.



	High Arousal	Low Arousal
Positive	Awe Excitement Amusement (Humor)	Contentment
Negative	Anger Anxiety	Sadness

Berger, J., & Milkman, K. L. (2012). *What makes online content viral?* J. of marketing research, 49(2), 192-205.

Tellis, G.J., MacInnis D.J., Tirunillai S., and Zhang Y. (2019) What Drives Virality (Sharing) of Online Digital Content? The Critical Role of Information, Emotion, and Brand Prominence, Journal of Marketing, 83 (4), 1-20

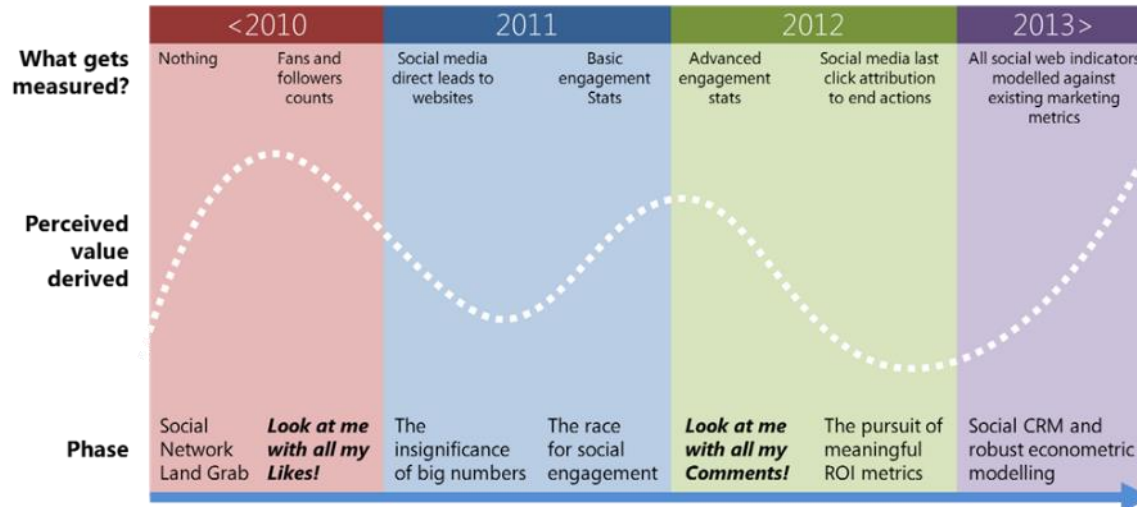
Social Media Metrics

Social Media Analytics: Measuring the Impact of SM Activity

Social Media Analytics: means a lot more than counting likes, shares etc:

Today there is a more sophisticated understanding of Social Media analytics/modeling

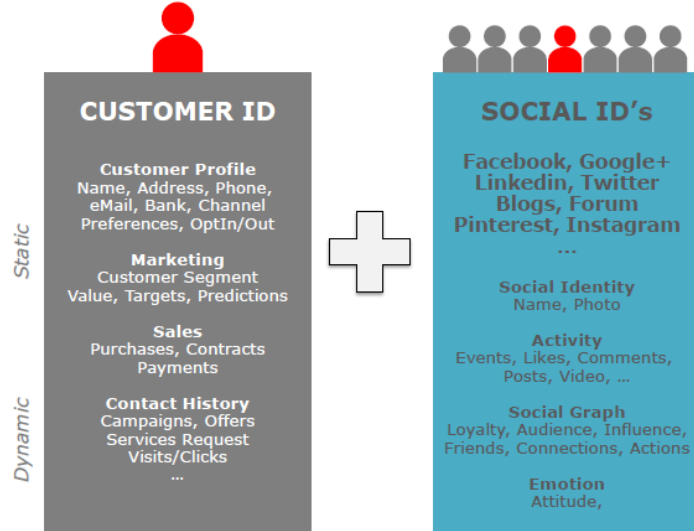
Finding answers to meaningful-valuable questions such as: *Do likes/shares have an impact on sales, conversions, engagement etc?*



Social Media Analytics: Integrating Customer ID + Social ID

Integrating Customer ID + Social Media activity (data)

Single Customer View = Ultimate Objective. However, this goal is hardly ever achieved: Privacy + Data Security Regulations



Simple Social Media Metrics

There is no recipe for must-use social media metrics. Depends on your marketing & campaign objectives.

Metric	Formula
Volume	The number of mentions of a brand name over a specified period of time. Often segmented into positive and negative volume using sentiment analysis.
Share of Voice (%)	$\frac{\text{Positive volume of brand}}{\text{Positive volume of all brands in category}} \times 100$
Engagement (per post)	The number of comments on, replies to, likes of, and shares of a given post.
Engagement (overall %)	$\frac{\text{Engagement at time } t \text{ with all posts to date}}{\text{Number of views at time } t \text{ of all posts to date}} \times 100$ or $\frac{\text{Engagement at time } t \text{ with all posts to date}}{\text{Number of followers at time } t} \times 100$
Advocates	The number of social media participants who write positive posts about a brand during a specified period of time.
Return on Investment (ROI)	$\frac{\text{Revenue from campaign} - \text{Cost of campaign}}{\text{Cost of campaign}} \times 100$
Leads Generated	The number of leads generated from social channels (sometimes expressed as a percent of all leads generated).
Response Time	The amount of time elapsed between the receipt of an inquiry or support request via social media and a response from the company.



* Twitter view counts show the amount of times a tweet has been viewed. They are essentially tweet impressions, meaning multiple views from the same account are included.

Social Media Metrics: What are Good Indicators?

IMPORTANT: The actual impact (ROI) of Social Media Engagement !

MOVING BEYOND a measurement of the simple number of fans/followers or “Like” metrics.

These metrics haven’t yet been proven to drive positive business results, whereas measures of **ACTUAL SHARING OF CONTENT** have been shown to deliver measurable marketing ROI. (Fulgoni, 2016)

BUYER'S JOURNEY	OBJECTIVE	SOCIAL MEDIA STRATEGY	SOCIAL ACTIVITY	SOCIAL KPI'S	BUSINESS IMPACT
AWARENESS	Create awareness	Expose target audience to brand content	Posts, promotions (boosts)	Impressions, reach	SOV, ToMA
CONSIDERATION	Generate demand	Generate engagement of target audience with brand content	Posts, responses	# of engagements, types of engagements	Visitors/traffic (online or offline)
DECISION	Drive conversion	Drive target audience to brand offers	Post, promotions	Link clicks	Conversions (purchases, lead submissions, app downloads, etc.)
ADOPTION	Delight customers	Drive engagement with brand product/services	Responses (i.e., social customer care)	(positive) earned mentions, customer care metrics (responses times & qty)	Sentiment and satisfaction
ADVOCACY	Inspire evangelism	Activate customer influencers	Posts, outreaches to influencers, reshares	Earned impressions, earned reach, Social UGC	Referrals, influencer activity, positive word of mouth, NPS

Simple SM Metrics: Share of Voice and Sales Effectiveness

Share of Voice: as a manager you may be mainly curious about (-) negative share of voice > Modify/Change the formula (adaptability)

Social Sales Effectiveness: Formula seems OK. But hey.... How can we possibly tell a purchase is coming from social channels?

The answer: Attribution Modeling (remember Lecture 1-2)

Share of Voice

$$\text{SOV} = \frac{\text{\# of conversations mentioning your brand}}{\text{total \# of industry conversations}}$$

% positive
% negative
% neutral



Sentiment Analysis

Social Sales Effectiveness

$$\text{Social Sales Effectiveness} = \frac{\text{\# of sales coming from social channels}}{\text{\# total sales}}$$

Growth in Sales from Social Channels:

- Month-over-Month
- Quarter-over-Quarter
- Year-over-Year

Simple SM Metrics: Conversation, Amplification and Applause Rates

Metrics such as (i) conversation rate, (ii) amplification rate and (iii) applause rate can be useful if aligned with what the marketer wants to learn.

	Twitter	Facebook	Google+	Blog	Youtube
# Posts:	8	9	4	3	4
# Comments:	13	22	5	22	5
# Re-tweets/Shares:	14	19	7	3	3
# Favorite clicks/Likes/+1:	3	81	19	16	28
1. Conversation Rate					
# Comments per post:	1.63	2.44	1.25	7.33	1.25
2. Amplification Rate					
# Re-tweets/Shares per post:	1.75	2.11	1.75	1.00	0.75
3. Applause Rate					
# Favorite clicks/Likes/+1 per post:	0.38	9.00	4.75	5.33	7.00
Imported data from analytics tool 1 time/day					
	Twitter	Facebook	Google+	Blog	Youtube
4. Economic Value:					
Value per visitor:	0.13 \$	3.72 \$	1.22 \$	1.43 \$	0.74 \$

Active engagement	 Conversation rate = $\frac{\text{\# of Comments/Replies}}{\text{\# of Posts}}$
	 Amplification rate = $\frac{\text{\# of Shares/Retweets}}{\text{\# of Posts}}$
	 Applause rate = $\frac{\text{\# of Likes/Favorites}}{\text{\# of Posts}}$
Active + passive (partly) engagement	 Economic value = $\frac{\text{Per Visit Goal Value from Google Analytics}}{\text{\# of Posts}}$

That Any Digital Marketing Manager May Wish to Have the Answer

those that can help in managing a social media page/account with right content, appeal and instruments. Here is a solid example:

The screenshot shows a Facebook post from the page 'Atrapalo Italia'. The post content includes a video titled 'MOVE' and a text update about restaurant menus in Barcelona. Red numbers 1 through 8 are placed around the interface, with lines pointing to specific elements: 1 points to the '4 checks in' status; 2 points to the video player; 3 points to the 'Move' video title; 4 points to the 'Like' button; 5 points to the '285 people like this' text; 6 points to the 'View all 42 comments' link; 7 points to the 'Write a comment...' text box; and 8 points to the 'Agosto sabe mejor: menús de día a 8 euros, y de noche a 19€' text. A green dashed line is drawn across the bottom of the image, separating the interface from the legend.

1 4 checks in

2 Y tú, después de verlo, ¿cómo te has quedado? ¿Vas a aguantar quieto este agosto? Busca entre nuestros viajes de último minuto y suéltate a partir de 19€ (26th día de 3) https://bit.ly.com/Viajes_3Dias_Minuto

3 Move agosto.com

4 Like

5 285 people like this.

6 View all 42 comments

7 Write a comment...

8 Agosto sabe mejor: menús de día a 8 euros, y de noche a 19€, con bebida y postre!

Hey, nos chivan que se están agitando las plazas ya, así que no te lo pases mucho y disfruta de los medodos y noches de agosto. :)

Agosto sabe mejor: menús de día a 8 euros, y de noche a 19€

1: Followers

2: Characters

3: Links

4: Videos or Images

5: Time

6: Date/Day

7: Likes

8: Comments



Popularity of Brand Posts on Facebook Fan Page

Main Question on Slide 1: *WHAT makes a Facebook (brand) post more POPULAR?*

Remember we need some X (predictors) and outcome/performance (Y) variables to answer this question and to build our model.

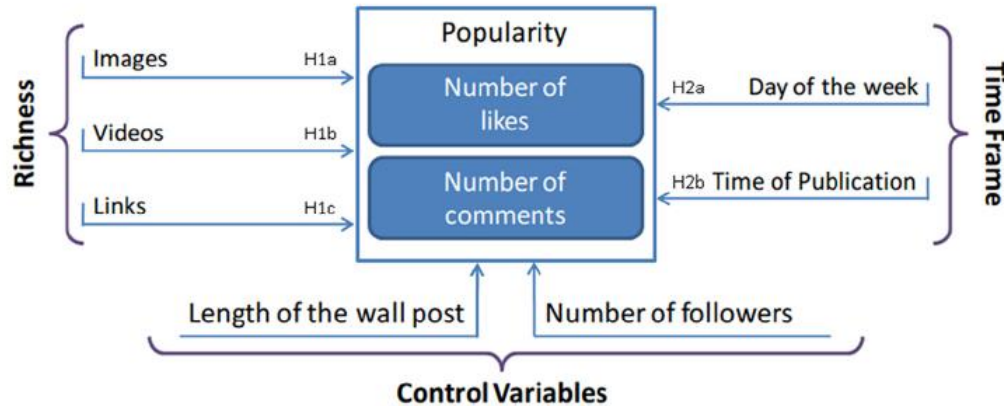
Now look at the Facebook page of the brand:

1 *What do we want to improve/make better ?*

These are your outcome Y variables

2 *What can we change to achieve this objective?*

These are your predictors X variables



Popularity of Brand Posts on Facebook Fan Page

This is how a researcher decompose and operationalize variables to use on a Facebook page for model based analysis.

Variable	Explanation	Codification and comments
Followers	Number of users that follow the travel agency Facebook fan page	Numerical ≥ 0 . Captured at the beginning of the data collection. It has been transformed applying natural logarithm function to better fit a normal distribution and improve the explanatory power of the model.
Characters	Post length measured by the number of characters	Numerical ≥ 0 . Characters of the links are also considered.
Links	Number of links within the post	Nominal-dichotomic 0: "no links" 1: "1 or more links"
Images	Number of images within the post	Nominal-dichotomic. 0: "no images" 1: "1 or more images"
Videos	Number of videos within the post	Nominal-dichotomic. 0: "no videos" 1: "1 or more videos"
Time	Publication time of the post	Nominal-dichotomic. 0: "non-business hours" (0:00-7:59 and 18:00-23:59 on Monday to Thursday; 0:00-7:59 and 15:00-23:59 on Friday; Saturday and Sunday at all hours) 1: "business hours (8:00-17:59 on Monday to Thursday; 8:00-14:59 on Friday)"
DateDay	Publication date of the post	Nominal-dichotomic. 0: "weekend" (from Friday at 15:00 to Sunday at 23:59) 1: "weekday" (the remaining time)
Likes	Number of likes that the post has got	Numerical ≥ 0 . It has been transformed using natural logarithm to better fit a normal distribution and improve the explanatory power of the model.
Comments	Number of comments that the post has	Numerical ≥ 0 . It has been transformed using natural logarithm to better fit a normal distribution and improve the explanatory power of the model.

Coefficients for significant variables in Likes Model.

	B	Std error	T	VIF
Videos	0.929	0.215	4.31 ^b	1.250
Images	0.673	0.145	4.65 ^b	1.205
LN(Followers)	0.632	0.089	7.11 ^b	1.189
Characters	0.003	0.001	5.09 ^b	1.370
(Constant)	-3.951	0.798	-4.95 ^b	

Coefficients for significant variables in Comments Model.

	B	Std error	T	VIF
Images	0.813	0.230	3.53 ^a	1.721
Time	0.651	0.227	2.87 ^a	1.033
LN(Followers)	0.293	0.112	2.62 ^a	1.076
Links	-0.627	0.205	-3.05 ^a	1.625
(Constant)	-1.320	1.032	-1.28	

Expected and obtained results by hypotheses.

Hypothesis	Expected	Likes Model	Comments Model
H1a (Images)	(+)	Supported	Supported
H1b (Videos)	(+)	Supported	Not supported
H1c (Links)	(+)	Not supported	Not supported (negative effect)
H2a (DateDay)	(+)	Not supported	Not supported
H2b (Time)	(+)	Not supported	Supported

Richness of the content (images/videos) raises the impact of the post in terms of likes.

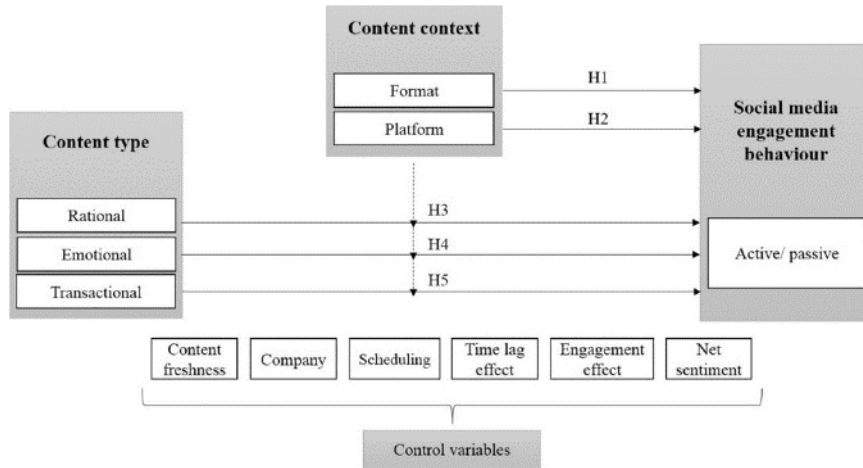
On the other hand, using images and a proper publication time are significantly influencing the number of comments,

Social Media Content and Its Impact on Customer Engagement

Here is another example with a **cause (driver) and effect question** from our reading list:

“What kind of social media content drives people’s engagement on social media : In other words: What kind of content should we post ? “

What **other factors** do also have an impact?: Timing? Product Type ? Platform (Facebook vs Twitter ?)



Now let’s think together

What could be our (Y) here – how can we measure engagement

What could be our (X) here – what/how can we change the content

Social Media Content and Its Impact on Customer Engagement

$$Y_{ij} = \alpha_{1j}\text{Informational} + \alpha_{2j}\text{Transformational} + \alpha_{3j}\text{Transactional} + \alpha_{4j}\text{Format} + \alpha_{5j}\text{Platform} + \alpha_{6j}\text{Freshness} + \alpha_{7j}\text{Company} + \alpha_{8j}\text{Userbase} + \alpha_{9j}\text{Scheduling} + \alpha_{10j}\text{Info} * \text{Frmt} + \alpha_{11j}\text{Inf} * \text{Plt} + \alpha_{12j}\text{Transf} * \text{Frmt} + \alpha_{13j}\text{Transf} * \text{Plt} + \alpha_{14j}\text{Trans} * \text{Plt} + \alpha_{15j}\text{Trans} * \text{Frmt} + \varepsilon_{1j} \quad (1)$$

Y = Dependent Variable

Model 1 Y = Number of Likes

Model 2 Y = Number of Comments

Two separate models for (i) likes and (ii) comments

X variables = predictors (explanatory variables)

			Number of likes		Number of comments	
			Without interaction effects model	With interaction effects model	Without interaction effects model	With interaction effects model
Independent variables	Adjusted R Square		0.752	0.763	0.572	0.605
	F		285.099***	196.808***	126.301***	93.937***
	(Constant)		1.593	1.731	-0.369	-0.498
	Format		0.263***	0.258***	-0.390***	-0.566***
	Platform		0.399***	0.250***	-0.712***	-0.405***
Controls	Rational		-0.131***	-0.239***	0.146***	0.260***
	Emotional		-0.025*	-0.005	0.017	-0.069*
	Transactional		-0.020	-0.082*	0.071	0.165***
	Freshness		0.022	-0.018	-0.319	-0.229*
	Company		-0.440***	-0.408***	0.281***	0.247***
	Scheduling		-0.051*	-0.050*	0.073	0.071*
	Time lag	Like lag	0.235***	0.209***	-	-
		Comment lag	-	-	0.113***	0.090**
	Net sentiment		0.999***	0.967***	-0.814***	-0.789***
	Engagement effect	No of likes	-	-	0.692***	0.733***
Interactions		No of comments	0.352***	0.374***	-	-
	Rational_format		-	0.105**	-	-0.123**
	Rational_platform		-	0.112***	-	-0.130**
	Emotional_format		-	-0.055*	-	0.132***
	Emotional_platform		-	0.029	-	-0.004
	Transactional_format		-	0.021	-	0.073
	Transactional_platform		-	0.069*	-	-0.258***

(Significance level of significant variables: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, - $p < 0.1$)

VIDEO format (coded as = 0) posts encourage users to sharing their opinion (**COMMENTS**)

PHOTO formatted (coded as=1) content stimulates passive users' engagement through **LIKES**.

Social Media Analytics: **Unstructured Data: Text, Photo and Video Data**

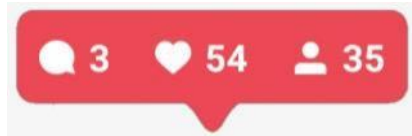
Unstructured Data: Text, Image & Video Data

Structured Data: We can add, sum, get averages and make mathematical statistical operations (*number* of likes, *number* of reshares etc)

TIP: "Keyword in structural data: number = quantity"

Unstructured Data: In text, photo or video format. Very common on Social Media and Review Platforms. Then how do we deal with it?

Structured Data



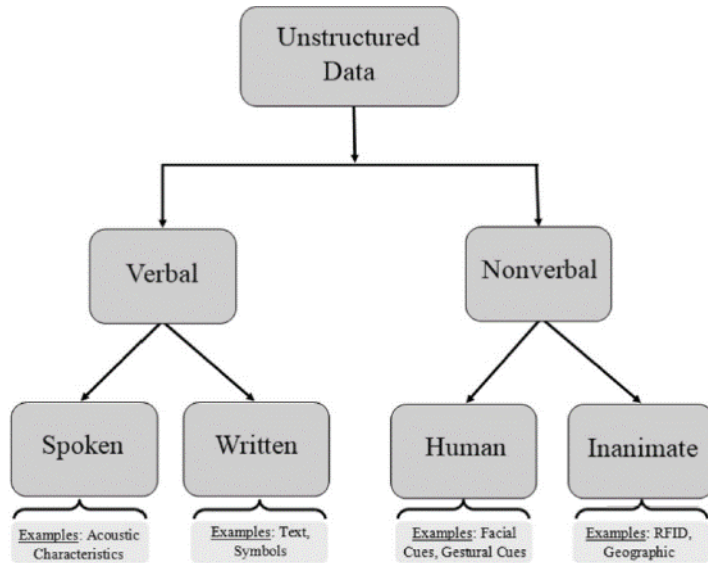
Unstructured Data



Unstructured Data: Text, Image & Video Data

Unstructured Data – Definition:

Data unit in which the information offers a concurrent representation of its multifaceted *nature without predefined organization or numeric values*.



Highly Unstructured Data: No predefined, numerical representations of the constructs of interest, so the researcher must “assign” numeric values to the data, either manually (e.g., human coding) or with automated methods (e.g., unsupervised machine learning) before analysis can take place.

Example: Highly unstructured data units are purely nonnumeric (e.g., video) requiring the researcher to assign values to the data. For example, a researcher interested in detecting customer satisfaction from UGC text data may choose to create a context-specific dictionary to determine the presence and degree of customer-expressed affect.

Highly Structured Data: Predefined, numerical representations of the constructs of interest; in instances when it does not (e.g., categories), the researcher can easily assign numeric values to the data.

Example: Extreme examples are purely numeric (e.g., net sales measured in USD), but other structured data units require just minimal researcher effort to translate the data into numeric values. For example, with surveys, the researcher determines the construct of interest (e.g., interfirm trust) and uses a scale on which participants indicate their degree of perceived interfirm trust, according to the given scale points. The participant’s response will not be strictly numeric, but the researcher can easily translate the responses into numeric representations of interfirm trust.

Unstructured Data: Text, Image & Video Data

1 **Non-Numeric:** Text, Photo or Video Data. To be converted into numeric data to be analyzed.

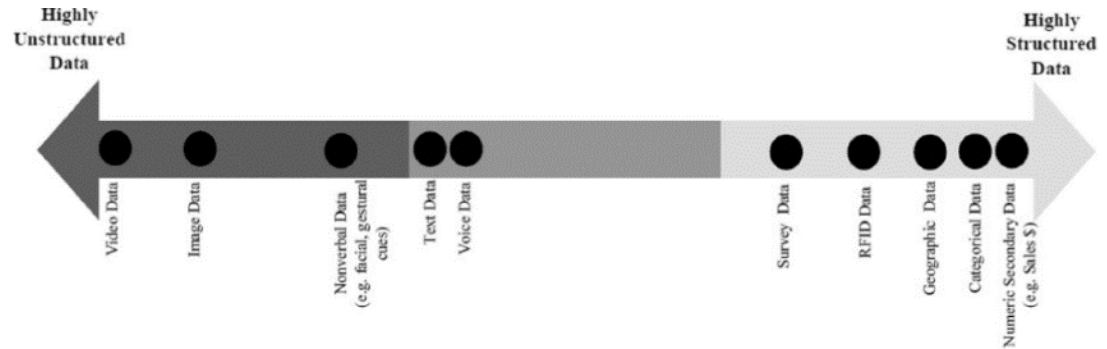
2 **Multifaceted:** Possesses multiple facets, each offering unique information.

Example: A photo = brightness, mode, contrast, colors or a Voice: pitch, speech rate, intensity

3 **Concurrent Representation:** Each facet represents a different phenomena at the same time.

Example: An online customer review has a positivity (valence) but also represents a topic (price, service, service etc)

 Text files and documents	 Server, website and application logs	 Sensor data	 Images
 Video files	 Audio files	 Emails	 Social media data



Source: Article 3.1 - Balducci, B., & Marinova, D. (2018). Unstructured data in marketing. Journal of the Academy of Marketing Science, 46(4), 557-590.

Source (Visual): <https://searchbusinessanalytics.techtarget.com/definition/unstructured-data>

Unstructured Data 1: Text Data on Online Platforms

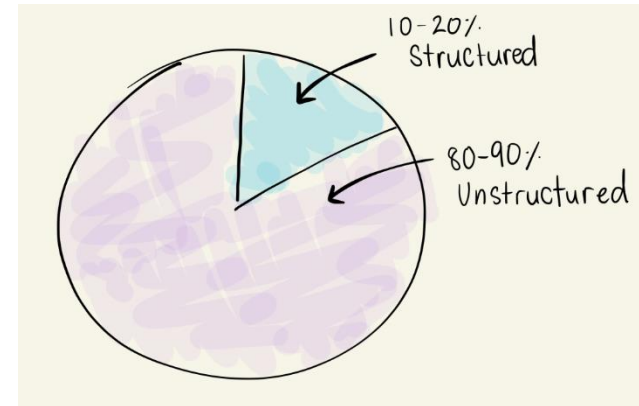
An estimated 80-90% of data held by firms today are unstructured *, and they are growing 15 times faster than structured data (SD)

Huge amount of text data (unstructured) is being generated every day. This goes way beyond social media posts and online reviews.

Some other examples: Song Lyrics, News Articles, Movie Scripts....

Textual Data Goes Beyond Social Media...

		Text Recipient		
		Consumers	Firms	Society
Text Generator	Consumers	Reviews, social Media	Call centers, chats	Online comments
	Firms	Product information, ads	Internal e-mails, financial reports	Financial reports, interviews
	Society	News, songs, movies	Business news (e.g., HBR)	Congressional discussions



Source: Netzer. O (2021) – Columbia Business School

Forbes (2022). Retrieved from <https://www.forbes.com/sites/forbescommunicationscouncil/2022/12/05/beyond-review-response-how-to-leverage-unstructured-data-to-improve-cx/?sh=51cfeee2c273>

Unstructured Data 1: Text Data Analysis > Classifiers & Extractors

分类器和提取器

1 Classifiers: group or tag data into a defined category (by sentiment, emotion, topic, etc.).

2 Extractors: retrieve pieces of information (like keywords, entities, phrases, numbers, etc.).

Sentiment Analysis is a sub domains of text-analytics based on the **classification of statements** being +/- (positive or negative)

CLASSIFIERS

I was **really happy** this morning walking into your store until when I needed to ask a question and **the staff were incredibly rude**. Normally you have **super fast service**. I wanted an item I saw online and tried to call the store but the **phone was always engaged**. Over experience was **not very nice**, the **staff didn't care**. Usually they are really **helpful**. I'm not sure what happened today.

EXTRACTORS

The screenshot displays a web browser window with the URL https://www.parfumsparis.nl/dames/4711/aqua-colonia-blood-orange-en-basil/eau-de-toilette-spray-170-ml_49386/. The page shows a product listing for 'Aqua Colonia Blood Orange & Basil' by 4711. The product is an Eau de Toilette Spray, 170 ml, with a price of € 41.40. The page includes a product image, a description, and a list of related products. The XML structure on the right side of the browser window shows the following elements:

- OV[1][class=page-cont]
- OV[2][class=page-cont]
- OV[3][class=product-review]
- OV[4][class=product-review]
- OV[5][class=product-review]
- OV[6][class=product-review]
- OV[7][class=product-review]
- OV[8][class=product-review]
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Unstructured Data 2: Images – Visual Data

Images are essential features of social media data. Images link the social media data with **rich contextual and consumption experiences**.

Classification & Clustering of social media images on certain dimensions is essential in their further analysis.

Example: Can I expect an increase in my branded (image) shares if I use more “glamorous” photos?



Unstructured Data 2: Images & Video – Visual Data

Some measures to be used in visual (video and photo) analytics:

Resolution, Aspect Ratio, Hue (warm vs cool colors), Brightness, Saturation, Contrast, Smoothness

Further: Facial Expression, Body Language, Event Detection (running, walking etc)



Example 1: Low difference: Distance between the above two image frames $d = 0.17608$.



Example 2: High difference: Distance between the above two image frames $d = 1.16550$.



(a) Low aesthetic score (0.14)



(b) High aesthetic score (0.58)

Figure 5: Example images with different aesthetic scores.

Unstructured Data 2: Images – Visual Data: Clustering Images

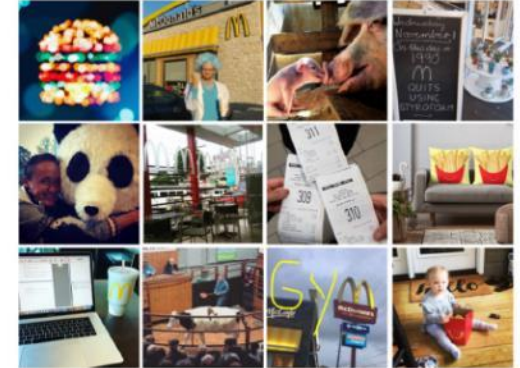
Example 1: Clustering of social media images based on different dimensions (dichotomies)

Example 2: McDonalds Social Data: Images are clustered (through visual processor) for further analysis.

This could be done either (i) manually – if you have little number of photos or (ii) visual data processing tools – for large data sets.

Note: If you do it manually (by search-copy-and human coding) – it is still OK. But this will take more time!

Figure 2. Sample Images of Brand Attributes and Their Antonyms in Flickr Data Set

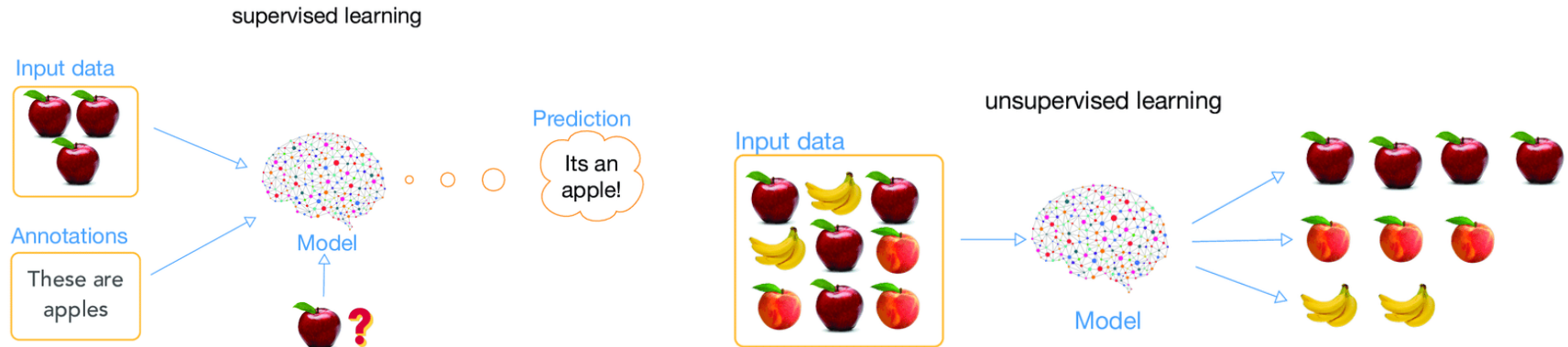


Unstructured Data 2: Images – Visual Data: Clustering Images

How to cluster images: different forms of machine learning: **(1) Supervised and (2) Unsupervised Learning**

Supervised learning, : model like ‘Teacher’ or ‘Supervisor’ who tells the machine the Label so that the machine knows which output from the given input.

Unsupervised Learning finds all kinds of unknown patterns in data.



Combining Textual and Visual Data

Data	Usage on Instagram	Example (see Fig. 2)	Insights
Data	Users upload images to document their everyday moments.	The image displays someone having food in a restaurant, holding twister fries.	What (e.g., objects) do the images display?
Text data	Users add a caption text to give further contextual information to an image. In doing so, they commonly reveal their emotional attitude towards the depicted content, facilitated by using emoticons to express their feelings.	The text expresses a positive attitude towards the product.	How do users feel about the situations?
Tag data	Users add tags to categorize and/or describe what is displayed in the image and to provide further contextual information associated with the image.	The tags describe the product at the center of the image and another product in background of the image, relate the post to the brand, and categorize the image as <i>foodporn</i> .	What do users associate with the situations?

Most impactful tags in the selected McDonald's image clusters.

Cluster	Most impactful negative tags			Most impactful positive tags	
Ice cream	theicecheckapp (-0.016)	regrann (-0.011)	justdelhiing (-0.006)	food (0.040)	mcflurry (0.053)
Burger	healthy (-0.007)	healthylifestyle (-0.003)	steakhouse (-0.002)	food (0.064)	burger (0.065)
Cartoon	trump (-0.009)	kfc (-0.007)	edgymemes (-0.006)	fanart (0.034)	followme (0.038)
Sundries	animalrights activism (-0.005)	deathrowdogs (-0.005)	vegetariansofig (-0.004)	govegan (0.024)	vegansofig (0.025)



TEXTUAL DATA (COMMENTS)

TAGS

VISUAL DATA (IMAGES)

Combining Textual and Visual Data

Sample process on how different components (visual, textual and numeric) data extracted and integrated in analytical process

Here; image-text similarity (consistence) analyzed with Machine Learning algorithm

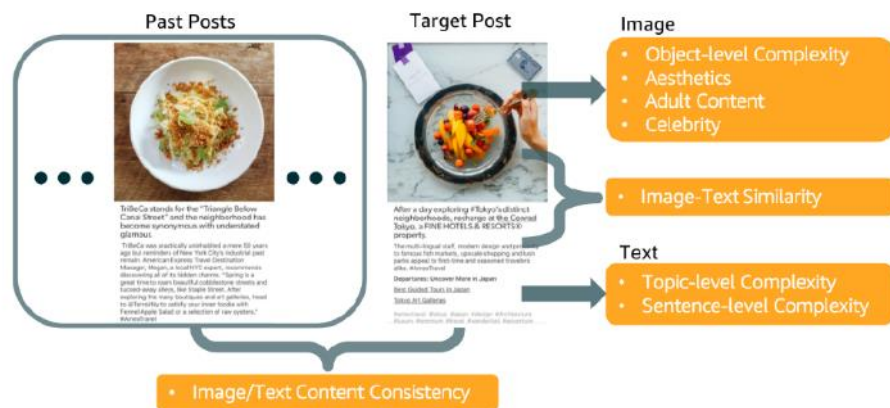


Table 4: Example posts and their image-text similarity scores.

Image			
Deep CNN output	quilt, comforter, comfort, couch, bed, window, bedroom, dorm, room, home, living room, bedspread, sofa, bed sheet, vintage, furniture, headboard	coffee mug, espresso cup, classroom, coffee shop, coffee, food, indoor, writing, beverage, hand, tea	jean, blue jean, denim, sweatshirt, closet, clothing store, craft, handkerchief
Text & Tags	Dream bed; interior, design, bedroom, bedding, comforter, apartment, dorm, home	Sunday; mani, coffee, rings, white nail polish, mug, laptop, Sunday, weekend	Find out how Forage Haberdashery and their bowties came into existence; dreamers, doors, bowtie, cloth, Forage Haberdashery
Image-Text Similarity	0.768	0.342	0

Example 1: Is a Picture Worth a Thousand Words? – The Impact of Image

Two managerially relevant questions (Article 3.3)

Are social media posts **with pictures** more popular than those without?

Why do pictures with **certain characteristics** induce higher engagement than some other pictures?

Table 2 shows how visual data is **coded (prepared)** = as being numeric data for modeling and analysis

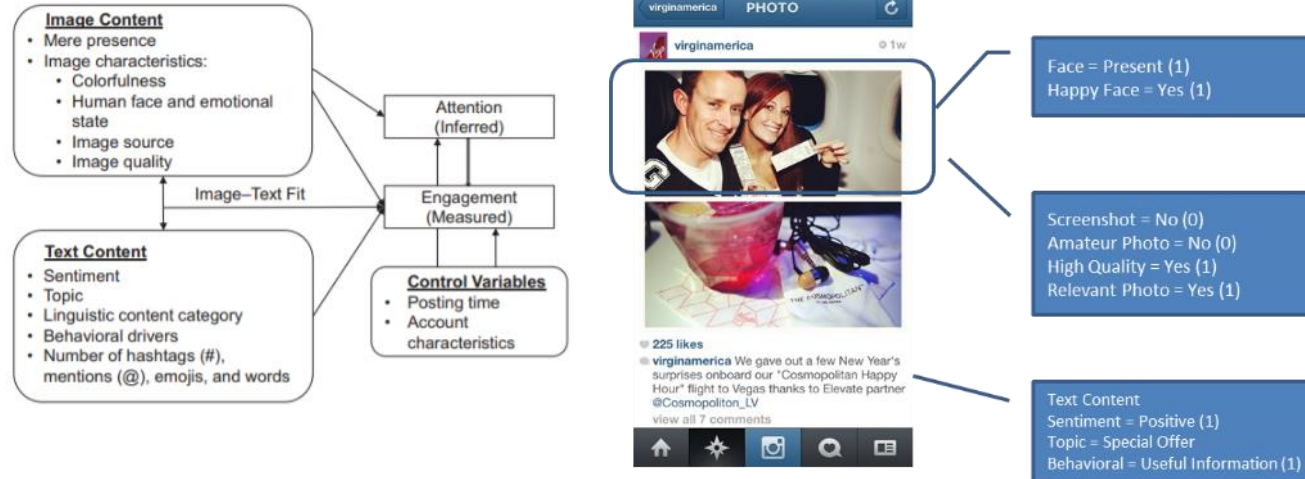


Table 2. Coding Categories.

Variables	Definition
Image Content	
Top 3 color percentage	Sum of pixel percentage of top-three colors. A smaller sum of percentage indicates a larger color variation.
Human Image	
Face presence	The image content is coded as 1 if at least one human face is detected; otherwise, it is coded as 0.
Happy face	The image content is coded as 1 if at least one happy human face is detected; otherwise, it is coded as 0.
Screenshot	The image content is coded as 1 if the picture is a screenshot of a web page; otherwise, it is coded as 0.
Amateur photo	The image content is coded as 1 if the picture is rated as a photo shot by an amateur photographer; otherwise, it is coded as 0.
High-quality picture	The image content is coded as 1 if the picture is rated as high quality; otherwise, it is coded as 0.
Relevant picture	The image content is coded as 1 if the picture is related to the topic of text component; otherwise, it is coded as 0.
Text Content	
Sentiment	The text content is coded as positive, negative, or neutral based on the overriding sentiment of the post.
Topics	The text content is classified as one of the six topics (different for each product category). The baseline is "Other Public Topic."
Behavioral drivers	Two dummy variables are created to indicate whether the text content of a tweet is shared to "provide useful information" (useful information) and "look good to others" (good image).

Example 1: Is a Picture Worth a Thousand Words? – The Impact of Image

(Mere) presence of an image has a positive impact on customer engagement on Twitter

High-quality and professionally shot pictures lead to higher engagement on Instagram and Twitter

Presence of human face and image–text fit can lead to higher user engagement on Twitter but not on Instagram.

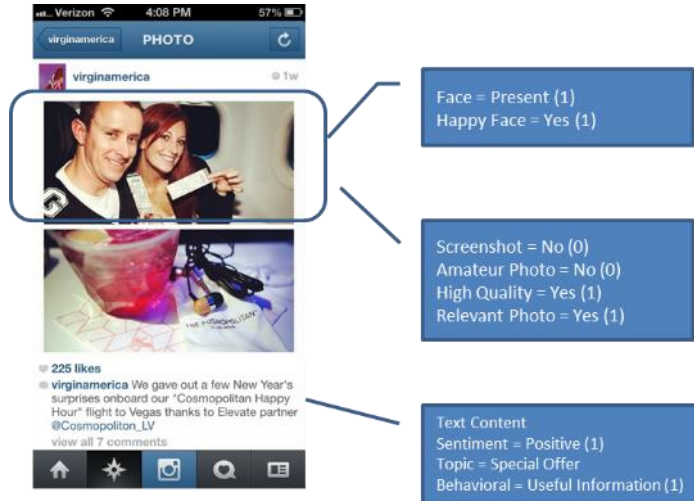


Table 8. Marginal Impact of Image Content.

	Airline Tweet		SUV Tweet		Airline Instagram
	Retweet	Like	Retweet	Like	Like
Direct image	+	+	+	+	N.A.
	(119.15%)	(87.26%)	(213.12%)	(151.56%)	
Linked image	–	–	N.A.	N.A.	N.A.
	(–78.18%)	(–66.20%)			
Top 3 color	–	n.s.	+	n.s.	–
	(–21.77%)		(26.89%)		(–4.50%)
Face presence	+	+	+	n.s.	n.s.
	(80.40%)	(38.76%)	(291.22%)		
Happy face	–	n.s.	–	n.s.	n.s.
	(–12.52%)		(–42.28%)		
Screenshot	–	n.s.	n.s.	–	–
	(–31.13%)			(–75.47%)	(–39.41%)
Amateur photo	–	n.s.	N.A.	N.A.	–
	(–24.78%)				(–11.97%)
High quality	+	n.s.	n.s.	+	+
	(20.30%)			(49.14%)	(9.89%)
Relevancy	n.s.	+	N.A.	N.A.	n.s.
		(42.45%)			

Notes: Cells with + or – symbols are significant at $p < .05$; n.s. = not significant; N.A. = not applicable.

Example 2: The Impact of Social Media Images on Consumer “Likes”

In this example, we again have an interesting and valuable question:

Which visual elements/factors in Social Media images (photos) of brands create more “likes” among customers/social media users ?



Fig. 1. Sample images of low, medium and high complexity for each individual measure of feature complexity.

FC = Feature Complexity, DC = Design Complexity

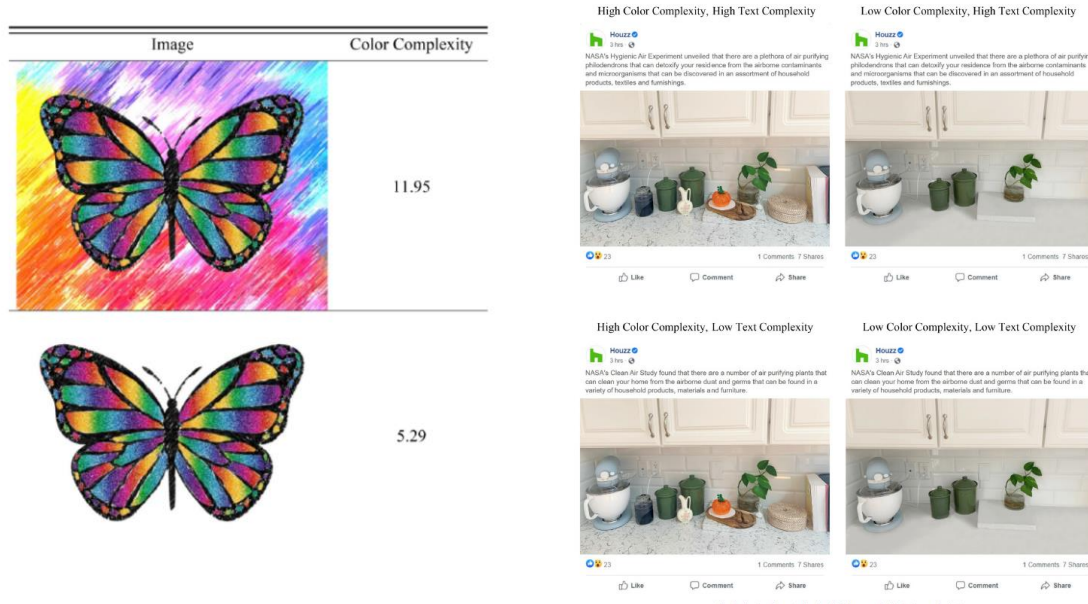
Type	Measure
FC	Color Complexity
FC	Edge Density
FC	Luminance Entropy
DC	Object Count
DC	Object Arrangement Asymmetry
DC	Object Arrangement Irregularity
DC - Control	Region Count
FC - Control	Frequency Factor
Content Control	Photography measures
Content Control	Face Detection
Content Control	Adjective-Noun Pairs
Content Control	Scenes

Example 3: The Impact of Color Complexity on User Engagement

Question: When and why image characteristics can draw social media users' attention and increase engagement?

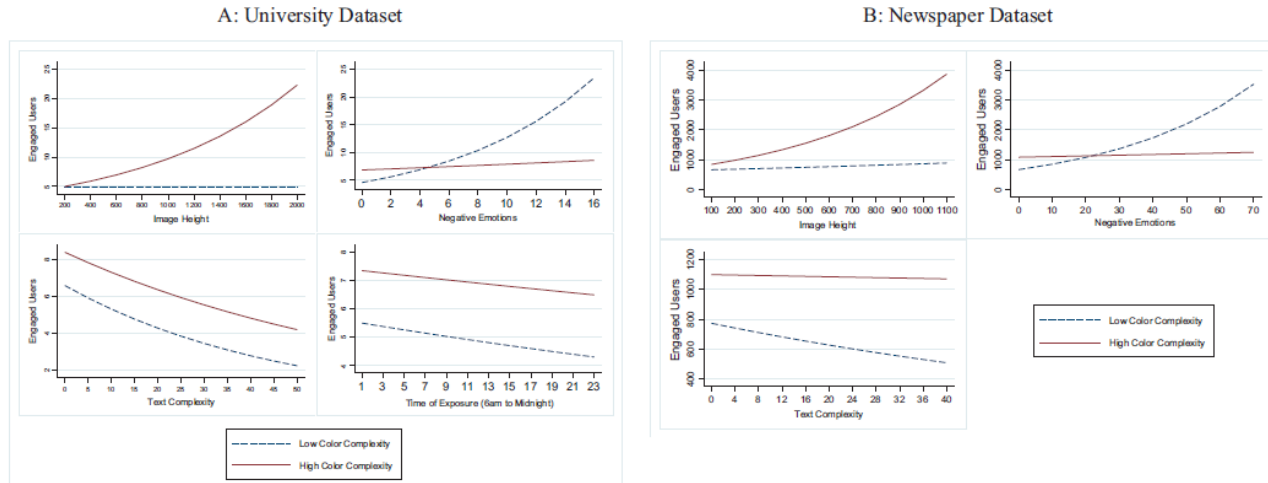
Color complexity in images can help in draws social media users' attention through shift from the peripheral to central processing mode.

This shift leads to **deeper processing of the social media post**, thus increasing the likelihood of the users' engagement with the post.



Example 3: The Impact of Color Complexity on User Engagement

- Images with **higher color complexity** receive **more attention**, thereby increasing the likelihood of user engagement with social media posts.
- **Image height** positively moderates the association between color complexity and user engagement.
- **Negative sentiment (anger, anxiety, fear) in the text** of an image **negatively moderates** the association between color complexity and engagement.
- **High text complexity** in the text of an image **negatively moderates** the association between color complexity and engagement.



Article 3.4: Kanuri, V. K., Hughes, C., & Hodges, B. T. (2024). Standing out from the crowd: When and why color complexity in social media images increases user engagement. *Int. Journal of Research in Marketing*, 41(2), 174-193.

Multimedia Data on Social Media = Multimodal Data

What is Dell multimodal approach?

Go to Dell account - Look at the latest posts from Dell.

Dell's social media strategy for multimodal data? Which posts are more effective?

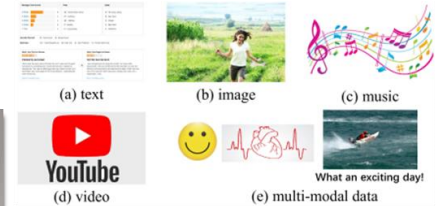
Dell account: <https://twitter.com/Dell>

Data Exercise

- Go to CANVAS – Modules – Exercise dell (.csv for Excel OR .sav for SPSS)
- Based on the dataset, which posts are more effective?
- Which modalities lead to more views? What about likes, replies and retweets?
- Hint: create dummy variables

Multimodal column:

1. Only text
2. Text + video
3. Text + photo
4. Only photo



Choose “only text” as reference group

Doubts about dummy, regression etc? Look at the slides from week 1



Thank You